Moment Interpolation and Recoverable Sewing for Honeycomb Bridge-Length Windows

A source-specific Ram analysis of cylinder amplitudes

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Abstract

We analyze the finite-volume and sewing interfaces in the September 26 manuscript *Cylinder amplitudes and logarithmic bridge-length windows on the honeycomb lattice*. Its all-fixed-order cut moments yield, by explicit interpolation, length moments of every order $p = 2 - 2/k$, $k \geq 2$. Keeping the exponents through pruning and recoverable exterior connectors gives narrower logarithmic windows, conditionally on the manuscript's physical polygon, flux, and connector estimates. We distinguish the auxiliary scale from the actual output height: every fixed symmetric height-relative exponent $1/12+\varepsilon$ is admissible after retuning the connector scale, but the endpoint $1/12$ and a fixed-height assertion do not follow. A separate uniform factorial-moment estimate reduces the recovery cutoff from $\sqrt{R}\log R$ to $C_A\sqrt{R\log R}$ for any prescribed polynomial error R^{-A}. Exact paired comparisons of the source's positive Fourier weights supply finite physical-norm certificates, not nonvanishing cylinder amplitudes by themselves. The global integrable and geometric inputs are identified rather than independently certified by finite checks. Contribution assessment: C3/5, the report proves a conditional narrower bridge-length window and a sharper recoverability cutoff with explicit scale and moment-order obligations, without a new unconditional self-avoiding-walk exponent theorem.

1 Physical model, status, and the increment

Let the honeycomb vertices be centers of unit equilateral triangles, with ports at side midpoints, as in the source [1]. The length $|\omega|$ is the number of visited centers, and

$$X = (2+\sqrt{2})^{-1/2}, \qquad \mu(\omega) = X^{|\omega|}.$$

The symbol μ denotes a positive, generally unnormalized weight, not a probability distribution. Joining compatible paths at ports multiplies their weights when their interiors are disjoint. A strict bridge in $\mathcal{B}_H$ starts at a fixed port of level zero, ends at any port of level H, and has all visited centers strictly between these levels. Lateral endpoints are not prescribed. We use the source's integral level units and its convention $B_0 = 1$. An arch has both endpoints on one wall and no intermediate wall contact; its positive endpoint gap is measured in the same lattice units. Changes between Euclidean and level distances cost fixed geometric constants, never a growing logarithm.

Plane polygons are unrooted modulo primitive translations. On a cylinder the source instead quotients winding polygons by whole-row translations, retaining column labels. Dividing again by

the circumference would change the normalization. Likewise its mixed-seam signs and the odd-size augmented Pfaffian are part of the model, not replaceable by an arbitrary positive determinant.

The source proves, on its stated route, a lower weight in the window

$$h^{4/3}(\log h)^{-1/8} \le |\omega| \le 2h^{4/3}(\log h)^{1/2}, \qquad 1 \le H \le 2h(\log h)^{1/64},$$

summed over heights, of order at least $h^{3/4}(\log h)^{-C}$. Our increment concerns the logarithms in this final deduction, not the power $4/3$ or its upstream proof. We use ordinary Hölder interpolation and factorial-moment estimates with their classical attribution. The source-specific content is keeping their exact orders, matching them to the two exterior connectors, and resolving the change from the auxiliary scale h to output height H.

The exact connective constant is an established theorem of Duminil-Copin and Smirnov [2]. The primary abstract of Krachun–Panagiotis [3] gives quantitative sub-ballisticity and polynomial decay of bridge weights, not the precise bridge exponent or length window used here. The September source is treated as a manuscript with a substantial claimed proof, not as an accepted settlement inferred from its existence. Reading it in full is not the same as independently verifying all of its uniform asymptotic and topological arguments.

2 Declared source inputs

Constants below are independent of large scales but can depend on fixed angles, confinement ratios, and a *fixed* moment order. Put $\mathfrak{b}(s) = 1 + \log\log(e^e + s)$, exactly the positive convention used in the source's final moment argument. All sums use physical lattice paths, not arbitrary point clouds.

Polygon and fixed-cut inputs. For polygons of diameter at most Cs, the source's Sections 9 and 11 give

$$Q_2(s) := \sum_P \mu(P)|P|^2 \le C\mathfrak{b}(s)s^{2/3}. \tag{1}$$

For each fixed integer $k \ge 1$ its double-cut count $M_g(P)$ satisfies

$$\sum_P \mu(P)M_g(P)^k \le C_k g^{-2+k/2}. \tag{2}$$

The count refers to eligible double crossings with gap comparable to g and within a fixed multiple of g of the bottom, in the source's polygon sewing construction. It is not the number of arbitrary pairs of vertices. Ordered distinct cuts, two outer caps, and pairs of strict bridges between consecutive cuts produce (2); expanding repeated cuts only changes C_k. No estimate on the growth of C_k is supplied here.

Let $F_{A,p}(s,g)$ be the length-p weighted arch mass with diameter at most s and gap in a fixed dyadic bin $[g,2g]$. Let $F_{B,p}(s,d)$ be the corresponding strict bridge mass of height d. Fixed changes of these cutoffs do not alter the following bounds. The source's injective or bounded-multiplicity sewings imply

$$cg^{-5/4}F_{A,p}(s,g) \le C \sum_{\operatorname{diam} P \le Cs} \mu(P)|P|^p M_g(P), \tag{3}$$

$$cs^{-3/2}\sum_{d\le s} d^{-1/4}F_{B,p}(s,d) \le C \sum_{\operatorname{diam} P \le Cs} \mu(P)|P|^p M_s(P)^2. \tag{4}$$

The same geometric maps apply to $p \ge 1$ since the input length is at most the completed polygon length. We import the actual path maps and their mass factors, not an abstract incidence-matrix proxy.

Deep-path and flux inputs. Average the source's deep arch/bridge measures over the integral heights $d \in [h, 2h]$, normalized by the number of such heights, and write $\bar{\mu}_h$. Deep means visiting the middle third of the strip. Sections 7, 10, and 11 provide

$$\int 1 \, \mathrm{d}\bar{\mu}_h \le Ch^{-1/4}, \qquad \int |\omega| \, \mathrm{d}\bar{\mu}_h \ge ch^{13/12}(\log h)^{-1/12}, \tag{5}$$

$$\bar{\mu}_h\{\operatorname{diam}\omega > t\} \le Ch^{-1/4}e^{-ct/h} \quad (t \ge C_0 h). \tag{6}$$

The first-moment lower bound uses the divided-slice positive gas and the signed boundary identity; replacing its negative inner exits by positive ones would not be legitimate.

Exterior growth and recovery inputs. Source Section 10 constructs length-controlled, recoverable growth from an initial scale to an exact target wall, losing a fixed power of their scale ratio and adding length at most a fixed multiple of the target scale to the power 4/3. Source Section 11 places two such paths on opposite sides of a divider below an arch. The left one uses two directions and a first qualifying cut. The geometry works for g/R and s/R tending to zero, with R/g at most a fixed power of $\log h$; its constants are fixed before h. The labels are the input arch, its placement, and the two actual connector paths, *not* all internal growth histories.

For the output class $\mathcal{G}_R$ of strict bridges of height R and diameter at most C_0R, let M count the double-crossing horizontal lines above both endpoints. The separately proved *uniform-order* estimate in source Section 11 is

$$M \le C_0 R, \qquad \int_{\mathcal{G}_R} \binom{M}{k} \, \mathrm{d}\mu \le C^k R^2 \big(C\sqrt{R/k}\big)^{k-1}, \quad 1 \le k \le c_0 R. \tag{7}$$

Here C, c_0, C_0 do not depend on k or R. This is a different input from (2); the latter cannot be used with k of order $\log R$.

3 The full fixed-order interpolation range

Theorem 3.1 (Length moments approaching order two)**.** *Assume* (1)–(4)*. For any fixed integer* $k_0 \ge 2$*, put* $p = 2 - 2/k_0$*. Then*

$$F_{A,p}(s,g) \le C_p \mathfrak{b}(s)^{p/2} s^{p/3} g^{p-1/4}, \tag{8}$$

$$\sum_{d \le s} d^{-1/4} F_{B,p}(s,d) \le C_p \mathfrak{b}(s)^{p/2} s^{1/2+4p/3}. \tag{9}$$

Consequently, for $t \ge h$ *(with fixed cutoff adjustments),*

$$\int_{\operatorname{diam}\omega \le t} |\omega|^p \, \mathrm{d}\bar{\mu}_h \le C_p \mathfrak{b}(t)^{p/2} (t/h)^{A_p} h^{4p/3-1/4}, \qquad A_p = \tfrac{1}{2} + \tfrac{4p}{3}. \tag{10}$$

Proof. Hölder with exponents $2/p$ and $k_0 = 2/(2-p)$ gives, on any diameter-restricted polygon set,

$$\sum \mu |P|^p M_g \le Q_2(s)^{p/2} \Big(\sum \mu M_g^{k_0} \Big)^{1/k_0} \le C_p \mathfrak{b}(s)^{p/2} s^{p/3} g^{p-3/2}.$$

Indeed $(-2 + k_0/2)/k_0 = p - 3/2$. Multiplying by $g^{5/4}$ proves (8). For (4), the second Hölder factor uses the fixed moment $2k_0$, and

$$\Big(\sum \mu M_s^{2k_0} \Big)^{1/k_0} \le C_p s^{(-2+k_0)/k_0} = C_p s^{p-1}.$$

After multiplication by $s^{3/2}$ this gives (9). Thus the bridge estimate requires twice the cut-moment order used by the arch estimate.

Sum (8) over dyadic gaps $g \le Ct$. Since $p - 1/4 > 0$, this costs only a constant and bounds the arch contribution by $C_p \mathfrak{b}(t)^{p/2} t^{4p/3-1/4}$. For bridges with $d \in [h, 2h]$, remove $d^{-1/4}$ at cost $Ch^{1/4}$ in (9), then divide by the order-h number of heights. The resulting bound is $C_p \mathfrak{b}(t)^{p/2} h^{-3/4} t^{1/2+4p/3}$, exactly the right side of (10). It also dominates the arch bound because $(t/h)^{3/4} \ge 1$. Averaging cannot increase the unrestricted arch mass. All rearrangements concern nonnegative sums. □

The source explicitly uses $p = 1, 3/2$. The theorem extracts the larger range already permitted by its all-fixed-order estimate; it does not claim a new Hölder inequality or a uniform $p \uparrow 2$ constant. In particular it does not establish a second length moment for the open paths.

4 Pruning with an explicit logarithmic budget

Write $\lambda = \log h$ for large h and fix parameters $a, r, u, v, \delta > 0$. Set

$$s = h\lambda^a, \quad R_0 = h\lambda^r, \quad \ell_- = h^{4/3}\lambda^{-v}, \quad \ell_+ = h^{4/3}\lambda^u, \quad g_{\min} = h\lambda^{-\delta}. \tag{11}$$

Proposition 4.1 (Four losses)**.** *Assume the preceding source inputs and Theorem 3.1 with a fixed $p > 1$. If*

$$v > \tfrac{1}{12}, \qquad u(p-1) > \tfrac{1}{12} + aA_p, \qquad \tfrac{3\delta}{4} - \tfrac{a}{3} > \tfrac{1}{12}, \tag{12}$$

then paths of diameter at most s, length in $[\ell_-, \ell_+]$, and (for arches) gap at least $g_{\min}$ retain first moment $\ge ch^{13/12}\lambda^{-1/12}$ and averaged mass $\ge ch^{-1/4}\lambda^{-u-1/12}$ for all sufficiently large h.

Proof. On a diameter bin $(t, 2t]$, Hölder, (6), and (10) bound its first moment by

$$C_p \mathfrak{b}(2t)^{1/2} h^{13/12} (t/h)^{A_p/p} \exp[-c(1 - 1/p)t/h].$$

The exponent of h is $-(1 - 1/p)/4 + (4p/3 - 1/4)/p = 13/12$. Summing $t = 2^j s$, $j \ge 0$, is $o(h^{13/12}\lambda^{-1/12})$ because $a > 0$ is fixed. This uses genuine exponential diameter control, not Markov's inequality with a merely bounded diameter moment.

At remaining diameters, $L 1_{\{L > \ell_+\}} \le \ell_+^{1-p} L^p$, where $L = |\omega|$. Thus the upper-length loss is

$$C_p \mathfrak{b}(s)^{p/2} h^{13/12} \lambda^{aA_p - u(p-1)}. \tag{13}$$

The lower-length loss is at most $\ell_- \int 1 \, \mathrm{d}\bar{\mu}_h \le Ch^{13/12}\lambda^{-v}$. Finally, (8) at $p = 1$ and a geometric sum over $g \le g_{\min}$ give the small-gap first moment

$$C\mathfrak{b}(s)^{1/2} s^{1/3} g_{\min}^{3/4} = C\mathfrak{b}(s)^{1/2} h^{13/12} \lambda^{a/3 - 3\delta/4}. \tag{14}$$

Every strict inequality in (12) absorbs the fixed powers of $\mathfrak{b}(s) = O(\log\log h)$. Subtract these four losses from (5) and divide the retained first moment by the upper length ℓ_+. □

The distinction between a first-moment lower bound and retained mass is essential: the final division by ℓ_+ is not optional. Nor does a small unweighted exceptional mass alone control a length-weighted loss.

5 Polynomial-error recovery at a smaller cutoff

Theorem 5.1 (Recoverability threshold)**.** *Assume (7). For each fixed $A > 0$ there is a fixed K_A such that, with $T_R = K_A\sqrt{R\log R}$,*

$$\int_{\mathcal{G}_R\cap\{M>T_R\}} M\,\mathrm{d}\mu \le C_A R^{-A} \tag{15}$$

for all sufficiently large R. The constants depend on those in (7); the assertion is not uniform in A.

Proof. For integers $1 \le k \le \min(c_0R, T/2)$ and $T \ge 2$, positivity, $M \le C_0R$, and monotonicity of the binomial coefficient give

$$\int_{\{M>T\}} M\,\mathrm{d}\mu \le \frac{C_0R}{\binom{\lfloor T\rfloor}{k}} \int \binom{M}{k}\,\mathrm{d}\mu.$$

For $n \ge k$, $\binom{n}{k} = \prod_{j=0}^{k-1}(n-j)/(k-j) \ge (n/k)^k$, since each factor is at least n/k. As $\lfloor T\rfloor \ge T/2$, absorbing the fixed constants in (7) gives

$$\int_{\{M>T\}} M\,\mathrm{d}\mu \le D_0R^{5/2}\sqrt{k}\left(\frac{D\sqrt{Rk}}{T}\right)^k. \tag{16}$$

Choose $\eta = (A+4)/\log 2$, $k = \lfloor \eta\log R\rfloor$, and $K_A = 2D\sqrt{\eta}$. For sufficiently large R, both constraints on k hold and the parenthesis is at most $1/2$. The right side is at most $2D_0R^{5/2}\sqrt{\eta\log R}\,R^{-(A+4)} \le C_AR^{-A}$. □

The source takes $T_R = \sqrt{R}\log R$ and k of order $(\log R)^2$, obtaining a superpolynomial tail. We require only a fixed polynomial tail to recover a polynomially large labelled mass. The smaller cutoff improves the multiplicity divisor by $(\log R)^{1/2}$, with constants depending on the selected error power. A single fixed K_A is not claimed to give all powers simultaneously. The factor $R^{5/2}$ and the unnormalized measure have been retained; applying a probability tail bound to μ without its mass would be an error.

6 Narrower windows and actual output heights

Theorem 6.1 (Conditional logarithmic parameter region)**.** *Assume all source inputs in Section 2. Fix $k_0 \ge 3$, $p = 2 - 2/k_0$, and positive a, r, u, v, δ satisfying (12) and*

$$a < r, \qquad \frac{4r}{3} < u. \tag{17}$$

Then, for some $c, C > 0$ and all sufficiently large h,

$$\sum_{h\le H\le 2R_0} \mu\{\omega \in \mathcal{B}_H : \ell_- \le |\omega| \le 2\ell_+\} \ge ch^{3/4}(\log h)^{-C}. \tag{18}$$

Rounding heights and scales to the admissible lattice changes only fixed constants. The contribution can be chosen in either $H \in [h, 2h]$ or $H \in [R_0, 2R_0]$. This is a height-summed statement, not a lower bound at every height in their union.

Proof. By Proposition 4.1, retained averaged mass is at least $ch^{-1/4}\lambda^{-u-1/12}$. If bridges carry at least half, multiply their averaged mass by the number of heights, which is comparable to h. This proves the stated lower bound in $[h, 2h]$.

Otherwise at least one strip contains retained arches of mass $a_h \geq ch^{-1/4}\lambda^{-u-1/12}$. Reflection and reversal restrict to one wall and gap direction at a fixed-factor cost. Put the common source on $y = 0$, with the arches above this line. For an integer $R \in [R_0, 2R_0]$ the source's two-connector construction applies: $s/R \to 0$, $g/R \to 0$, and $R/g \leq C\lambda^{r+\delta}$. All scale-ratio costs are therefore fixed powers of λ. This is where $a < r$, positive fixed parameters, and the exact-target version of growth are needed.

For completeness, the right connector stays below $y = 0$ on its side of the divider and grows to the right target wall. The left connector first grows along the oblique u direction to a terminal bin at order-R distance. Extracting its first qualifying u-cut sums over order-R targets but divides by at most an order-$R^{3/4}$ suffix weight. The extracted prefixes therefore retain a logarithmic mass factor. A second growth, strictly beyond that cut, reaches the exact left wall. Fixed ray tolerances are chosen before h so that this second part remains below the arch and away from the divider. These are the source's geometric assertions, included among our premises; no finite moment certificate creates the disjoint paths.

There are order-R placements and each actual connector has weight at least $R^{-1/4}\lambda^{-C}$ and length at most $CR^{4/3}$. Port multiplicativity gives labelled output mass

$$W_R \geq cR^{1/2}\lambda^{-C}a_h. \tag{19}$$

In an output, selecting a double-crossing y-line recovers the arch and its two endpoints, hence its placement, gap, and exterior paths. The first qualifying u-cut is unchanged by the second left stage, which stays strictly beyond it. Thus the inverse multiplicity is at most CM, not a product of independent numbers of growth histories.

By Theorem 5.1, the labelled mass with $M > T_R$ is at most $C_A R^{-A}$. Taking, for example, $A = 2$ makes this negligible relative to (19), whose lower bound is $h^{1/4}$ times a fixed logarithmic power. Dividing the remaining labelled mass by CT_R yields

$$\mu\{\text{retained outputs at height } R\} \geq c\, a_h\, \lambda^{-C}(\log R)^{-1/2} \geq ch^{-1/4}\lambda^{-C'}.$$

The two connector lengths total at most $CR^{4/3} \leq C'h^{4/3}\lambda^{4r/3} = o(\ell_+)$, whereas the arch length is at least ℓ_-. Hence all counted outputs have length in $[\ell_-, 2\ell_+]$ for large h. Sum over order-R_0 heights and rotate to the fixed bridge orientation. This proves (18). □

6.1 Two explicit interpretations

The following rational example improves the source's upper logarithm in the *auxiliary* scale without changing its height cutoff:

$$k_0 = 8, \quad p = \tfrac{7}{4}, \quad a = \tfrac{1}{512}, \quad r = \tfrac{1}{64}, \quad u = v = \tfrac{1}{8}, \quad \delta = \tfrac{1}{4}.$$

Here $A_p = 17/6$. The positive margins are

	$v - 1/12$	$u(p-1) - 1/12 - aA_p$	$3\delta/4 - a/3 - 1/12$	$u - 4r/3$	$r - a$
value	$1/24$	$5/1024$	$53/512$	$5/48$	$7/512.$

Thus $2h^{4/3}(\log h)^{1/8}$ replaces the source's $2h^{4/3}(\log h)^{1/2}$, retaining its lower cutoff and $R_0 = h(\log h)^{1/64}$. It would be incorrect to call this a symmetric H-relative exponent $1/8$: in the connector branch the lower exponent becomes $v + 4r/3 = 7/48$.

Corollary 6.2 (Height-relative symmetric windows)**.** *Under the same source inputs, fix $\varepsilon > 0$ and put $e = 1/12 + \varepsilon$. There are fixed $r, c, C > 0$ such that, for all sufficiently large h,*

$$\sum_{h \le H \le 2h(\log h)^r} \mu\left\{\omega \in \mathcal{B}_H : H^{4/3}(\log H)^{-e} \le |\omega| \le H^{4/3}(\log H)^{e}\right\} \ge ch^{3/4}(\log h)^{-C}. \tag{20}$$

The contributing heights can be restricted to one of the two bands in Theorem 6.1; the selected band may depend on h.

Proof. A fully constructive parameter choice is

$$v = \tfrac{1}{12} + \tfrac{\varepsilon}{4}, \quad u = \tfrac{1}{12} + \tfrac{\varepsilon}{2}, \quad r = \min(\varepsilon/32, 1/100), \quad a = \min(r/2, \varepsilon/128), \quad \delta = \tfrac{1}{4} + a.$$

Define $D = \varepsilon/2 - 19a/6 > 0$, and choose an integer $k_0 \ge 3$ with $k_0 > 2u/D$. Since $A_p \le 19/6$ and $p - 1 = 1 - 2/k_0$,

$$u(p-1) - \tfrac{1}{12} - aA_p \ge D - \frac{2u}{k_0} > 0.$$

The other pruning and connector inequalities are immediate: $r > a$, $4r/3 < u$, $v > 1/12$, and $3\delta/4 - a/3 = 3/16 + 5a/12 > 1/12$. In addition,

$$u < e, \qquad v + \frac{4r}{3} \le \frac{1}{12} + \frac{\varepsilon}{4} + \frac{\varepsilon}{24} < e.$$

For $h \le H \le 2h(\log h)^r$, one has $\log H / \log h \to 1$ uniformly. The lower target length is at most a fixed constant times $h^{4/3}(\log h)^{4r/3-e}$, which is $o(\ell_-)$ by the last strict inequality. The upper target length is at least a fixed constant times $h^{4/3}(\log h)^{e}$, which dominates $2\ell_+$ because $u < e$. Thus all outputs counted in (18) satisfy the height-relative window for sufficiently large h. □

This is not a fixed-length counting asymptotic, a fixed-height lower bound for every H, or a limit law for typical bridges. The endpoint $e = 1/12$ is inaccessible to this argument: even the short-length loss then need not be negligible, and a fixed $p < 2$ leaves no upper-tail margin as $a \downarrow 0$. Letting ε depend on h would also make k_0 vary, requiring new control of C_{k_0} in (2).

7 Exact paired physical Fourier certificates

The window proof uses positive path sums. A separate useful comparison concerns the Fourier measures underlying the source's Pfaffian and determinant amplitudes. Put $k = 2 + \sqrt{2}$ and $c = 3/4$. With removable values at $\nu = 0$, source Section 4 uses

$$\mu(\nu) = \frac{\nu \cosh(\pi\nu/4)}{\sinh(\pi\nu)}, \qquad \mu_0(\nu) = \frac{\nu}{2\sinh(c\pi\nu)},$$
$$w_b(\nu) = \frac{k}{\sqrt{2}} \frac{\sinh(\pi\nu/4)}{\sinh(\pi\nu)}, \qquad w_0(\nu) = \frac{k}{2\sqrt{2}\cosh(c\pi\nu)}.$$

Here $\mu(\nu)$ is a density, distinct from the path weight $\mu(\omega)$; its argument prevents ambiguity. All four densities are strictly positive on the real axis, even, exponentially decaying, and have every polynomial moment.

Proposition 7.1 (One common contrast, opposite signs)**.** *For every real ν,*

$$\mu = \mu_0(1 + \Delta), \qquad w_b = w_0(1 - \Delta), \qquad \Delta(\nu) = \frac{1}{2\cosh(\pi\nu/2)} \in (0, 1/2]. \tag{21}$$

Consequently, on every finite polynomial space, with the physical complex inner product $\langle f, g\rangle_w = \int \overline{f} g\, w$,

$$G_{\mu_0} \preceq G_\mu \preceq \tfrac{3}{2} G_{\mu_0}, \qquad \tfrac{1}{2} G_{w_0} \preceq G_{w_b} \preceq G_{w_0}. \tag{22}$$

These inequalities have constants independent of polynomial degree.

Proof. Put $x = \pi\nu/4$. The addition formulas give

$$2\cosh x \sinh 3x = \sinh 4x + \sinh 2x,$$
$$2\sinh x \cosh 3x = \sinh 4x - \sinh 2x.$$

Since $\sinh 4x = 2\sinh 2x \cosh 2x$, division gives (21) away from zero and continuity gives it at zero. Multiplying its scalar inequalities by $|f(\nu)|^2$ and integrating proves the matrix inequalities in any common coefficient basis. Strict positivity and polynomial integrability justify the inner products and guarantee invertible finite Gram matrices. □

For example, if $h_j(w)$ is the squared norm of the monic degree-j orthogonal polynomial, minimization over all monic polynomials yields

$$\tfrac{1}{2} h_j(w_0) \le h_j(w_b) \le h_j(w_0), \qquad h_j(\mu_0) \le h_j(\mu) \le \tfrac{3}{2} h_j(\mu_0).$$

For a basis $\phi_0, \dots, \phi_n$, write its evaluation row as $E(z) = (\phi_0(z), \dots, \phi_n(z))$. The exact finite evaluation norm is $K_w(z) = E(z) G_w^{-1} E(z)^*$, including at nonreal z. Therefore

$$K_{w_0}(z) \le K_{w_b}(z) \le 2K_{w_0}(z). \tag{23}$$

This explicitly uses the complex conjugate transpose, not a formal bilinear square of a complex evaluation vector.

Proposition 7.2 (Physical residual interval)**.** *In a* w_0*-orthonormal polynomial basis let* $G = G_{w_b}$ *and let* a *be any coefficient vector. Put* $r = E(z)^* - Ga$ *and* $J(a) = 2\,\mathrm{Re}(E(z)a) - a^*Ga$. *Then*

$$J(a) + \|r\|_2^2 \le K_{w_b}(z) \le J(a) + 2\,\|r\|_2^2\,. \tag{24}$$

Proof. Completing the square gives $K_{w_b}(z) - J(a) = r^* G^{-1} r$. By (22), $I \preceq G^{-1} \preceq 2I$. □

This is a classical residual identity made quantitative by the source's exact weight ratio. It requires exact entries or certified entry intervals and physical basis normalization. It says nothing about the rounding error of an unspecified quadrature. Moreover determinant comparison multiplies the norm constants across all degrees: a factor $2^{-(n+1)}$ can remain. Uniform finite norm equivalence alone does not prove a nonzero cylinder amplitude after the source's leading exponential terms cancel.

7.1 A normalization example and a failure test

The source's transform identities at zero, or direct elementary hyperbolic integration, give

$$\int w_b = 1, \qquad \int w_0 = \frac{2(1+\sqrt{2})}{3}, \qquad \int \mu = 2 - \sqrt{2}, \qquad \int \mu_0 = \frac{4}{9}.$$

For w_b the first identity is the value at zero of $k/(2\cosh s + \sqrt{2})$; for μ it is the derivative at zero of $2\sinh s/(2\cosh s + \sqrt{2})$. The reference identities follow from the transforms of sech and $\nu/\sinh(c\pi\nu)$,

or their integrable geometric expansions. Thus even at degree zero the two reference measures do not have the same mass as their physical measures. In particular

$$K_{w_b} = 1, \qquad K_{w_0} = \frac{3}{2(1+\sqrt{2})}.$$

The inequalities in (23) hold, whereas silently normalizing both reference and physical weights to mass one would change the comparison constants and the determinant normalization.

Here is a general counterexample to a sampled substitute. Given finitely many real quadrature nodes $\nu_1, \dots, \nu_m$, take $P(\nu) = \prod_{j=1}^{m}(\nu - \nu_j)$. Every nonnegative quadrature sum of $|P|^2$ at these nodes is zero, but $\int |P|^2 w_b > 0$. Therefore finite-node positivity is not a physical-norm certificate on all polynomial degrees. At a chosen finite degree, one still needs a proven quadrature error and unisolvence, neither of which follows from merely obtaining a positive numerical matrix.

8 What the original Ram proofs contribute

Three original local sources were freshly inspected, rather than using a later catalogue description as a substitute for proof. Their role here is analytic bookkeeping, not ownership of classical Hölder, Fourier, or Gram identities.

The original TR250 passages [5] formulate finite polynomial moment coordinates and the physical weighted norm. If $p = \sum_j a_j \phi_j$ and $G_{ij} = \int \overline{\phi_i}\phi_j w$, then $\|p\|_w^2 = a^* G a$. Under an invertible coordinate change $a = Tb$, the physical matrix becomes T^*GT, rather than G or the identity. The positivity proof integrates the squared modulus of a nonzero polynomial against a genuinely positive weight. These are precisely the issues in Propositions 7.1 and 7.2.

The original physical-metric chapter [6] proves orthogonal projection and its residual accounting. If P_V is the physical orthogonal projection onto a finite-dimensional space,

$$\|f - p\|_w^2 = \|f - P_V f\|_w^2 + \|P_V f - p\|_w^2 \qquad (p \in V).$$

Consequently a small finite residual only controls the second term. In the cylinder calculation, the omitted-degree or asymptotic remainder remains a separate obligation. The chapter's distinction between a finite Gram matrix and a continuum frame directly prevents promotion of (24) to a cylinder-amplitude theorem.

The original Fourier chapter [7] establishes inversion under explicit integrability assumptions, Plancherel on its proper L^2 domain, moment differentiation under domination, and convolution with the required Fubini conditions. The present densities have exponential real-axis tails, so every fixed polynomial moment is absolutely integrable. This justifies our finite moment manipulations; it does not justify arbitrary complex evaluation as an infinite-dimensional L^2 functional. The source's open-strip Hermite residual and imaginary evaluation must remain a finite polynomial calculation with its own bounds.

The meaningful connection is thus exact physical coordinates and error separation. The main logarithmic improvement in this report comes from the actual honeycomb sewing estimates, not from renaming them as a Ram frame.

9 Whole-source architecture and retained obligations

The following map covers every scientific module. *Imported* means the report uses or records the result as a source assertion without claiming a separate complete audit of its proof. All listed modules were literally read in full.

Module	Mechanism	Boundary retained here
Introduction	Physical ports, cylinder amplitudes, finite estimates and window	Lengths count visited centers. Height averaging, free terminal ports, and row-translation quotients are unchanged.
01	Local relations, vacuum uniqueness, polynomial interpolation, Pfaffian/determinant formula	The odd augmentation, seam signs, ordering, and actual row-invariant vacuum cannot be replaced by an isomorphic formal expression.
02	Charged contractions, zero-axis neutrality, root layers, coercivity and stability	The bosonic pairing is not globally positive in every charge. Uniform mode and root estimates remain imported.
03	Long-circle approximation and localized coupled roots	Error bounds allow the required source counts. Fixed per-root bases and reserved stability are not inferred from pointwise convergence.
04	Positive Fourier weights, orthogonal polynomials, amplitudes	Section 7 checks exact weight comparisons. Leading-term cancellations and nonzero amplitude asymptotics remain source inputs.
05	Pressure, sparse edge roots, weighted displacement control, positive clusters and cylinder tails	A finite positive inverse does not imply a uniform pressure bound. Winding and height-placement normalization matter.
06	Open-strip vacua, wall reflection, Hermite residual, bridge mass	The first surviving finite polynomial term and its uniform asymptotics are imported; arbitrary infinite L^2 evaluation is not asserted.
07	Two-mark divided gas, positive calibration, boundary flux	Loop fugacity and signed inner exits are preserved. Positivity of a different gas would not prove the chord first moment.
08	Two ports on the same polygon, neutral sliding	The two marks are not independent samples. This step supports the polygon moment route.
09	Oblique two-scale comparison, good pairs and width bins	Errors uniform in expansion order and calibrated physical scales are imported. They produce (1).
10	Positive caps, exact-target recoverable growth, confinement	Actual nonintersection and length-controlled growth are essential to the two exterior connectors. The later renewal tail is not needed for our refinement.
11	Aspect cutoff, fixed-cut moments, pruning, connectors, uniform recovery	Sections 3–6 independently prove the conditional quantitative extraction, keeping the two different moment-order quantifiers.

For orientation, the finite periodic representation uses

$$h(x,y)=x^2+y^2+\sqrt{2}xy,\quad F=\prod_{i<j}\frac{h(z_i,z_j)}{z_i-z_j},\quad Q=F\,\mathrm{Pf}_{\mathrm{ext}}\left[\frac{z_i^2-z_j^2}{h(z_i,z_j)}\right],$$

and $R=F^2\det[kz_iz_j/h(z_i,z_j)]$ in the source's conventions. Its cylinder quantity is built from $(k-1)^NQ^2/R$. At coincident parameters these are interpreted by the source's polynomial continuation, not naive division by zero. Showing a finite Gram representation for one positive specialization is only one piece of the route from these expressions to physical cylinder amplitudes and then to (5).

10 Costs, limitations, and research directions

What improves. Theorem 3.1 exposes all admissible fixed orders and retains the smaller logarithmic factor $\mathfrak{b}^{p/2}$. Theorem 6.1 replaces ad hoc logarithmic choices by a checkable parameter region.

Corollary 6.2 pays the actual output-height conversion rather than hiding it. Theorem 5.1 spends only the recovery accuracy needed for the labelled-mass lower bound. No algorithmic speedup or measured runtime is claimed.

What costs more. Small ε requires a large but fixed cut-moment order, twice that order on the bridge side, and a possibly enormous starting scale. The construction gives no effective universal numerical h_0 because the upstream constants are unspecified. The final mass still loses an unspecified fixed logarithmic power. Narrowing the length interval does not turn its weighted mass into a normalized probability without control of the denominator.

What cannot be removed by finite certificates. Actual signs, boundary contacts, path disjointness, and recoverable cuts remain physical statements. A list of admissible small polygons does not certify them at all scales. Positive Fourier densities control finite real-axis norms but not an asymptotic amplitude after cancellation. The source's arbitrary-accuracy long-circle and two-scale estimates carry uniformity obligations that are not consequences of our residual interval.

Feasible extensions with missing inputs named. An endpoint 1/12 result would need a sharper deep first-moment lower bound, smaller exceptional losses, or a controlled near-second moment beyond this fixed-order argument. An h-dependent ε would additionally require explicit growth bounds for C_k and compatible geometric constants. A theorem at every exact height would need a quantitative height-transfer or monotonicity argument for the length-restricted mass, not just for unrestricted bridge mass. Explicit norm-comparison and quadrature-error constants could support certified finite amplitude calculations; to reach an asymptotic amplitude they must be combined with the source's uniform remainder and noncancellation estimates. These are research obligations, not conclusions of the present report.

11 Evidence scope and current public context

The owned read ledger records pre-read hashes, literal numbered display intervals, post-read hashes, and unread complements. All 19 cached public text files were read: 4,476 lines, consisting of the 12 scientific modules, main file, bibliography, README, and four small figure sources. This includes the complete 431-line final sewing argument. Figure source was read as text; no claim of a separately compiled source-paper figure inspection is made. This is full textual coverage, not independent global proof certification.

Fresh private original proof reads total 533 lines: TR250 lines 121–180 and 425–477; the physical-metric chapter lines 1–210 in full; and the Fourier chapter lines 1–210. The remaining TR250 intervals and Fourier lines 211–373 are unread. The Fourier chapter's included subchapters and the physical-metric chapter's external figure were not inspected. There is no whole-corpus reading claim.

The public refresh on October 9, 2026 inspected primary abstracts and metadata for [2, 3], and the public family-237 formalization-scope page [4]. The latter describes selected summability, free-energy, strip-mass and displacement results, including order-$H^{-1/4}$ bridge mass support. It does not describe a verification of this report's narrower logarithmic window; no Lean files were run or independently audited here. Public documentation is also distinct from the captured Git-tree byte-identity record for the cached manuscript. No private material was submitted in literature queries.

The owned executable checks pass 2,972 parameter cases and 10,241 counted assertions, using exact rational arithmetic for interpolation exponents, constructive parameter margins, binomial

tail inequalities, the paired hyperbolic identities after rational exponential substitution, and finite positive contrast residual identities. Their receipt separates parameter cases and assertions. They are reproducible regression fixtures, not enumeration of all honeycomb paths and not certification of the imported source inputs. Independent external checks are not included in the owned counts.

No source formula is declared erroneous by this report. The source's choice to discard $\mathfrak{b}^{p/2}$ in favor of $\mathfrak{b}$, its selected moment orders, and its stronger recovery tail are valid but nonoptimal for this final inference. We correct neither the source's global theorem nor its topology by finite algebra; we improve the specified conditional deduction and state its limits explicitly.

12 Conclusion

The exact point of leverage is the combination of all-fixed-order polygon cut moments with a separate uniform-order recovery bound. The former narrows length tails, the latter reduces the recovery divisor, and the connector geometry transfers retained arches to actual bridges. Paying the auxiliary-to-output scale conversion gives the height-relative $1/12 + \varepsilon$ result with honest height-summed quantifiers. The physical Fourier comparisons give supporting finite certificates with exact domains; they do not replace the global amplitude mechanism. This combination earns a notable local, conditional contribution, not a claim of historical novelty for its classical tools or an independent proof of the full honeycomb exponent program.

References

[1] *Cylinder amplitudes and logarithmic bridge-length windows on the honeycomb lattice*, September 26, 2026. Cached manuscript in the OpenAI mathematics collection, family 237, sections 1–11 and introduction; the complete 19-text scoped subtree was literally read. Public manuscript directory.

[2] H. Duminil-Copin and S. Smirnov, *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$, Annals of Mathematics 175 (2012), 1653–1665. Publisher abstract and publication record. Public abstract and metadata checked; the external proof was not independently audited here.

[3] D. Krachun and C. Panagiotis, *Quantitative sub-ballisticity of self-avoiding walk on the hexagonal lattice*, arXiv:2310.17299v1, October 26, 2023. Primary abstract and version record checked, not the full proof.

[4] OpenAI mathematics collection, *The three-quarter exponent for honeycomb self-avoiding walk*, family-237 formalization scope documentation, accessed October 9, 2026. Scope description only; no independent Lean audit.

[5] M. Subbarao, original private `TR-rev-1-old-tr250.tex`, RamFoundationTRs. Fresh proof passages: lines 121–180 and 425–477. Finite moment coordinates, physical polynomial norms and complex-adjoint bookkeeping; confidential local source.

[6] M. Subbarao, RamCalBooks, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Original physical Gram, projection and residual proofs; confidential local source. External figure not inspected.

[7] M. Subbarao, RamCalBooks, volume 1 revision 2, `ch07-ramcs-rev2.tex`, lines 1–210. Original Fourier inversion, integrability, Plancherel and moment arguments; confidential local source. Remaining text and included subchapters not read for this report.